

# Predicting Pre-meal Glucose using Sparse Optimization and Time-Series features

Group Members: Syed Ahmad Shah,  
Christopher Ognibene

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# Meet the team

Both team members each contributed to the project in each developing independent optimization solutions and combining both approaches together in a compare and contrast discussion of the pros and cons based on accuracy metrics. Discussions of results – each member discussed their approach, and reviewed each other’s discussions for reasonableness and accuracy. Although we focused on two separate implementations, we have repeatedly met to discuss findings and alter our future works based on our results and progress.

## Syed Ahmad Shah Implementation 1

1. Data Extraction, Transformation, and Cleaning
2. Feature Engineering from glucose, insulin, meal, exercise, and time-series event data
3. Objective Function Formulation using regularized regression models
4. Implementation of Ridge, LASSO, Elastic Net, and HistGradientBoostingRegressor
5. Time-based train/test evaluation and comparison against baseline models
6. Patient-specific modeling, residual modeling, and feature-importance interpretation
7. Visualization, result analysis, and discussion of dataset limitations and future improvements

## Christopher Implementation 2

1. Data Preparation: extracted source data and transformed into wide format, with Codes as columns, and each row indexed by Patient ID-Datetime. Filled missing values with the mean of the code per patient-day.
2. Feature engineering: Averaged four pre-meal glucose measurements
3. Target Variable: computed a 1-day forward variable, per patient).
4. Objective function: formulated a traditional mean squared error function, with L1 and L2 penalty constraints.
5. Numerical solution: programmed key optimization functions by hand with custom functions for gradient descent, L1 and L2 penalties, and elastic net objective functions. This approach serves to provide a custom version of the optimization that can be tweaked and iterated in future.
6. Model evaluation: computed MSE, RMSE, and MAPE metrics, as well as examined residual plots and QQ-plots to assess fit.

# What is our Goal?

The **goal** of the **project** is to build a **glucose prediction model** that uses the **fewest meaningful predictors** to accurately predict glucose levels keeping prediction error low. In other words, we can examine the tradeoff between accuracy of **prediction vs. sparsity** (fewest predictors). A good model should have a sufficient balance that does not overfit the training data to the extent that test error is significantly worse (e.g., worse than random guess 50%).

The value of our project is to provide alternative toolkits to medical professionals working with patients with diabetes to continuously monitor and understand expected ranges of blood sugar levels pre, and post meals and throughout the day. Accurate recording of blood glucose levels at predictable intervals throughout the day can help a patient and their health care provider achieve the best possible outcomes to managing patient health and ensuring risk of adverse events (e.g., hyperglycemia) are minimized. In addition, there is a predictive benefit to understanding the correlation of patient blood glucose levels with activity indicators in helping providers understand diabetes risk factors and estimated glucose levels for new patients.

In summary, our project seeks to add a perspective to the health community in providing a way to pinpoint as accurately as possible the key potential drivers behind a particular patients glucose levels based on their recorded daily activity and inform patients distinguish helpful from unhelpful activities to managing blood glucose levels.

# Optimization Model

We explored two different implementations, described below.

## Implementation 1

We predict a patient's pre-meal glucose level using prior insulin, meal, exercise, and previous glucose levels, while selecting a sparse set of predictors using a linear model:

$$y_i = \beta_0 + \mathbf{x}_i^T \boldsymbol{\beta}$$

$y_i$  = glucose value for target observation  $i$

$\mathbf{x}_i$  = feature vector for observation  $i$

$\boldsymbol{\beta}$  = model coefficients

Where the shared error term between the tested Ridge, LASSO, and Elastic Net is the MSE:

$$\frac{1}{2n} \sum_{i=1}^n (y_i - \beta_0 - \mathbf{x}_i^T \boldsymbol{\beta})^2$$

In this implementation, the model is written in penalized form rather than an explicit constrained form. However, an existing constraint, is that our model's regression coefficients cannot grow arbitrarily large, as this may lead to overfitting, additionally we encourage the model to use a smaller subset of predictors that contribute the most towards the predictions

## Implementation 2

We predict a patient's next pre-meal glucose using current insulin dosage, post-meal glucose, and exercise activity, similarly identifying the most relevant predictors that minimize out-of-sample test error.

The objective function is a conventional least squares, with additional L1 and L2 regularization penalty functions to increase sparsity and prioritize features with greatest influence (i.e., largest drivers behind minimizing prediction error). Prediction error is defined as the difference between the predicted next-period pre-meal glucose level from the actual average next-period pre-meal glucose level of a patient, taken at a specific day based on a specific activity. We are careful to avoid data leakage by ensuring no next-period features are present in our training set.

Least squares is typically sensitive to outlier data, therefore, regularization can help to create a more robust solution by identifying the most impactful features on model prediction (i.e., largest coefficients). While there are no explicit constraints in the formulation of the objective function, the constraints are the additional terms appended onto the core MSE term in the L1, L2, and Elastic Net formulas.

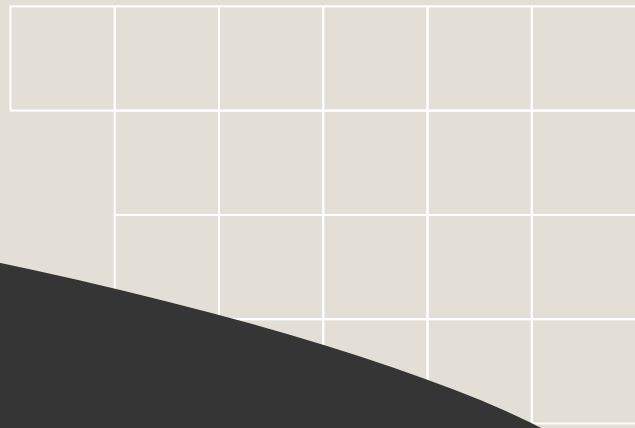
$$\text{MSE (unconstrained)} \min_{\mathbf{w}} f(\mathbf{w}) = \frac{1}{2n} \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{w} - y_i)^2 = \frac{1}{2n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$$

$$\text{L1 regularized: } \min_{\mathbf{w}} f(\mathbf{w}) = \frac{1}{2n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda \sum_{j=1}^p c_j |w_j|$$

$$\text{L2 regularized: } \min_{\mathbf{w}} f(\mathbf{w}) = \frac{1}{2n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \frac{\rho}{2} \|\mathbf{w}\|_2^2$$

$$\text{Elastic Net: } \min_{\mathbf{w}} f(\mathbf{w}) = \frac{1}{2n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda \sum_{j=1}^p c_j |w_j| + \frac{\rho}{2} \|\mathbf{w}\|_2^2$$

# Proposed Solution



# Implementation 1

## Is it convex/non-Convex

For the models implemented:

1. Ridge:
  - a. Uses a squared error term, which is convex
2. Lasso:
  - a. L1 penalty, which is convex
3. Elastic Net
  - a. L2 penalty, which is convex

As our models are convex, they won't have bad local minimas as do non-convex models, thus our solver will be more reliable.

## What is the feasible set

The feasible set depends strictly on how we define to go about the problem. In the code, we use the sklearn library, which uses the penalized form: minimized prediction error + penalty.

In that way the feasible set are all possible coefficient vectors, with a penalty that discourages large or too many coefficients.

For

1. Ridge this means the square coefficient size is bounded
2. LASSO the total absolute coefficient is bounded
3. Elastic Net this means both the square coefficient size and total absolute coefficient is bounded

## Why did we conclude a solution exists?

A solution exists as the objective function is continuous and regularized, and the feasible set is also convex and bounded. Additionally, the penalties prevent the coefficients from growing too large, thus the optimization problem is well constrained.

## Numerical Solution using a Toolbox

From the library “scikit-learn” we utilized:

1. RidgeCV
2. LassoCV
3. ElasticNetCV
4. HistGradientBoostingRegressor  
(For comparison)

When implementing Ridge, sklearn solves the convex regression, and tunes the parameters through cross validation. For LASSO and Elastic Net, sklearn utilizes coordinate descent to update one coefficient at each iteration.

## Closed Form Solution ?

The models differ in regards to the closed form.

As Ridge Regression is a convex quadratic function, and by taking the derivative of the objective function with respect to its coefficients and setting it to 0, we can obtain a solvable system of linear equations

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

However, LASSO and Elastic Net do not have closed form solution as they include an L1 penalty, which introduces a non-differentiable term when evaluated at zero.

# Implementation 2

The next solution's approach employs a least-squares objective function formulation, where the goal minimizes the prediction error between the vector product between our data and ideal model coefficients. We incorporate L1 and L2 regularization penalty functions, which are embedded constraints on the inclusion/exclusion and size of coefficients, aiming to achieve sparsity in the model (prioritize features with greatest driver for minimizing prediction error).

We consider the following:

- Prediction error is defined as the predicted glucose level from the actual average glucose level of a patient, taken at a specific day based on a specific activity.
- Reason: This was because we will ultimately run a regularized Linear Regression algorithm, which is sensitive to outliers. Regularized least squares is robust to outliers in the underlying data, and can help identify the most impactful features in the model with the highest coefficients.
- We aim to achieve a best fit surface but not perfect accuracy to minimize total error while maintaining sparsity (parsimonious) and out-of-sample prediction power. (i.e., achieve close enough estimation on future unseen patient activity data).





## Convexity

The objective function is convex. Since least squares is quadratic, we have a clear minimum, therefore, with a sufficient initial guess and step size, our algorithm is guaranteed to converge.

## Why did we conclude a solution exists?

The objective function is bounded below by zero and unbounded above, therefore a global minimum exists.

## Closed Form Solution ?

A solution exists as Least squares is convex, and can be solved analytically with the normal equations.

$$\hat{\beta} = (X^T X)^{-1} X^T \bar{y}.$$

However, we will use numerical solutions as we apply L1, L2, and elastic net regularization, where L1 penalty does not have a closed form solution (i.e., non-differentiable at  $x=0$ ).

## Numerical Solution using a Toolbox

For unconstrained least squares and L2 regularized least-squares, traditional gradient descent is used.

$$\beta_j \leftarrow \beta_j - \alpha \nabla g$$

For L1 regularization, we use proximal gradient descent which caps the weight update at 0 if the step is within  $\alpha * \lambda$  (i.e., near non-differentiable points ( $x=0$ )).

$$\text{prox}_{\lambda}(\beta_i) = \begin{cases} \beta_i - \lambda & \text{if } \beta_i > \lambda \\ \beta_i + \lambda & \text{if } \beta_i < -\lambda \\ 0 & \text{otherwise} \end{cases}$$

# Validating Solution

## About the Dataset:

We acquired the dataset from the **UC Irvine Machine Learning Repository** at the link here, <https://archive.ics.uci.edu/dataset/34/diabetes>. This was an open-source dataset provided for initial use for academic researchers and medical professionals, and made public to students or practitioners of machine learning for academic or personal project use. **The data contains details on basic physiology and patho-physiology of diabetes mellitus and its treatment. The dataset is composed of 70 diabetes patient records collected from electronic and paper records. Patients are indexed by 1-70.** Each patient has several weeks' to months' worth of outpatient care data.

### The feature names are:

**Data-Codes** ('Code'): a listing of the codes used in the data sets.

**Date** in MM-DD-YYYY format: i.e., date of recording

**Time** in XX:YY format: i.e., time of recording

**Value:** i.e., blood glucose level

Codes will be the main set of features we will use in our optimization problem approaches, as our goal is to predict glucose levels with respect to the key codes.

### Several key features are defined below:

**33 = Regular insulin dose** - short-acting insulin (works soon after injection, often around meals)

**34 = NPH insulin dose** - intermediate-acting insulin (lasts longer than regular, and is used for background insulin plan)

**35 = UltraLente insulin dose** - Long-acting insulin (Different role than regular insulin, as its not for immediate spikes)

**48, 57 = Unspecified blood glucose measurement** - Glucose was measured, but not tied to meal context

## Our Solution:

For both solutions, the following was the process followed to calibrate each of our objective functions to minimize prediction error for glucose level.

- **Data Extraction:** We extracted the original source data from the compressed zip file into individual CSV files (one per patient) using Python's data compression and analysis libraries. We aggregated the individual CSV files into one data set.
- **Data Transformation:** data is converted from 'long' to 'wide' format into the more traditional data set: the Code column was expanded into individual columns per code, corresponding to one recording (i.e., patient-timestamp).
- **Define the target:** Use rows with glucose codes 58, 60, 62, 64 (Pre-glucose measurements), as this is our objective.
  - **Implementation 1:** Utilizes historical features
  - **Implementation 2:** average of pre-glucose measurement codes per timestamp, then lag by patient to create a forward-looking target. Fill remaining N/A values with the mode. (i.e., most occurring measurement)
- **Build features:** Use only events that happened before each target measurement.
- **Clean the data:** Handle malformed dates, irregular times, and nonnumeric glucose values if needed.
- **Split by time:** Train on earlier observations and test on later observations.
- **Compare models:**
  - **Implementation 1:** candidate models Mean predictor, Ridge regression, LASSO, Elastic net, PCA baseline
  - **Implementation 2:** candidate models Unconstrained LS, L1 regularized, L2 regularized, and Elastic Net
- **Evaluate:** MSE, RMSE, Number of selected predictors, tradeoff between sparsity and accuracy

# Implementation 1

## Data Processing and Feature Engineering:

- The data was separated by **patient** and included the **date, time, code, and value**.
- To better fit the data for a regression model, the date was **normalized** to be “**Days since start**”, and Time to “**MinuteOfDay**” as the model would be better able to relate to continuous data as opposed to utilizing OneHotEncoding.
- Prior activity, insulin, and exercise indicators were engineered (shown right), which were used to train and calibrate the optimization models with prior activity as predictors of current pre-glucose levels.
- For each patient:
  - If Code = Target Glucose Measurement, form a unique observation by Code, Minute
  - Engineered features spanned prior 6, 8, and 24 hours from current pre-glucose measurement (i.e., do not equal current time to avoid data leakage)
- Total of 10,346 rows out of 58,661 raw rows.

| Feature                  | Lookback Window                                                    | Type          |
|--------------------------|--------------------------------------------------------------------|---------------|
| typical_meal_6h          | number of prior "66" events over last 6 hours                      | Diet          |
| more_meal_6h             | number of prior "67" events over last 6 hours                      | Diet          |
| less_meal_6h             | number of prior "68" events over last 6 hours                      | Diet          |
| regular_insulin_8h       | Sum of Measurements of "33" over last 8 hours                      | Dosage        |
| nph_insulin_24h          | sum of measurements of "34" over last 24 hours                     | Dosage        |
| ultralente_insulin_24h   | sum of measurement of "35" over last 24 hours                      | Dosage        |
| typical_exercise_6h      | number of prior "69" events over last 6 hours                      | Exercise      |
| more_exercise_6h         | number of prior "70" events over last 6 hours                      | Exercise      |
| less_exercise_6h         | number of prior "71" events over last 6 hours                      | Exercise      |
| prev_glucose             | Most recent pre or post glucose measurement                        | Glucose Level |
| hours_since_prev_glucose | Number of hours elapsed since last pre or post glucose measurement | Glucose Level |
| hypo_symptoms_24h        | number of past "65" events over last 24 hours                      | Glucose Level |
| glucose                  | current level of pre-glucose [the target]                          | Glucose Level |
| special_event_24h        | number of past "72" events over last 24 hours                      | Unspecified   |

# Implementation 1

- **Target definition:** We created one-hot encoded target variables for each of our four pre-glucose measurement features, because these will be our decision variables (1/0), with our actual target glucose level as stored in the 'Value' field.
- **Train/Test split:** we use the first 80% of rows stratified by patient for training, and the remainder for testing
- **Baseline 1: simple mean predictor** - Predict glucose using the average training glucose. This gives a weak baseline to beat (a "naive" prediction)
- **Baseline 2: previous glucose predictor** - Predict the next pre-meal glucose using the most recent previous glucose.
- **Ridge regression** - Fit ridge regression on all engineered features. Purpose: closed-form mathematical baseline, handles correlated predictors, good comparison point.
- **LASSO regression** - Fit LASSO on the same feature set. Purpose: sparse feature selection, convex optimization model, easier interpretation.
- **Elastic net regression** - Fit elastic net on the same feature set. Purpose: combines LASSO sparsity with ridge stability usually better when predictors are correlated likely your best main model.
- 5-fold cross validation was used, along with null value imputation using the median, and standard scaling (Z-score)

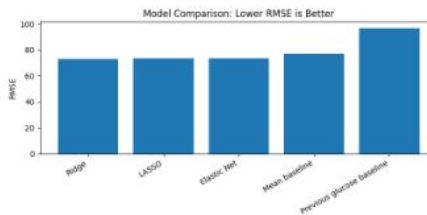


Figure: Comparison of Model Performance

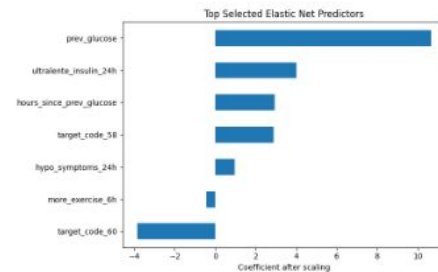


Figure: Feature Importance

|   | Model                     | MSE         | RMSE      | MAE       | Selected Predictors |
|---|---------------------------|-------------|-----------|-----------|---------------------|
| 2 | Ridge                     | 5332.484106 | 73.023860 | 58.338130 | 21.0                |
| 3 | LASSO                     | 5397.191415 | 73.465580 | 59.063308 | 7.0                 |
| 4 | Elastic Net               | 5413.498230 | 73.576479 | 59.236969 | 7.0                 |
| 0 | Mean baseline             | 5892.664434 | 76.763692 | 62.582119 | NaN                 |
| 1 | Previous glucose baseline | 9377.593643 | 96.837976 | 71.344254 | NaN                 |

# Implementation 1

- We conducted the same analysis as described on the prior slide, however, with the key difference of including **patient id as a feature** (i.e., one-hot encoded patient id as columns). Therefore, our decision variables include not just the 4 pre-glucose binary variables, but also **70 binary patient variables**.
- **Train/Test split:** rather than splitting by patient id strata, we split by patient based on the DAYS feature, with the first 80% of recordings for training, the remainder for testing.
- The five models are fit as described on the prior slide.

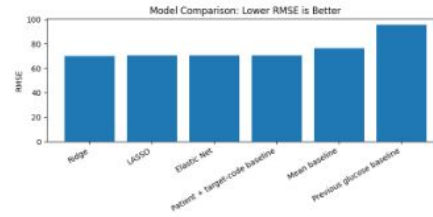


Figure: Comparison of Model Performance (with Patient ID and Code as Predictors)

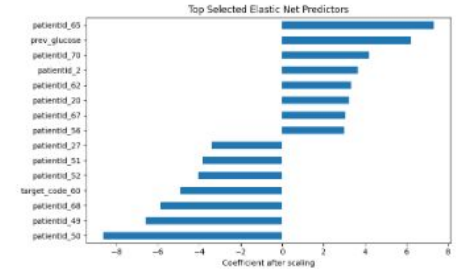


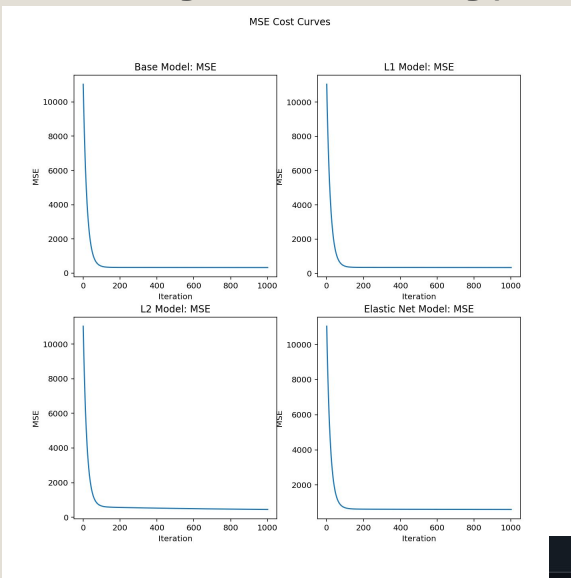
Figure: Feature Importance

|   | Model                          | MSE         | RMSE      | MAE       | Selected Predictors |
|---|--------------------------------|-------------|-----------|-----------|---------------------|
| 3 | Ridge                          | 4924.237908 | 70.172914 | 55.358211 | 86.0                |
| 4 | LASSO                          | 4952.979558 | 70.377408 | 55.741738 | 43.0                |
| 5 | Elastic Net                    | 4952.979558 | 70.377408 | 55.741738 | 43.0                |
| 2 | Patient + target-code baseline | 4954.792381 | 70.390286 | 54.691903 | NaN                 |
| 0 | Mean baseline                  | 5891.123437 | 76.753654 | 62.505170 | NaN                 |
| 1 | Previous glucose baseline      | 9194.173502 | 95.886253 | 70.567823 | NaN                 |

# Implementation 2

Learning Curves: as we can see, all methods converge with respect to a max iteration of 1000. Examining the performance metrics provides further insight to the differing precision of the algorithms.

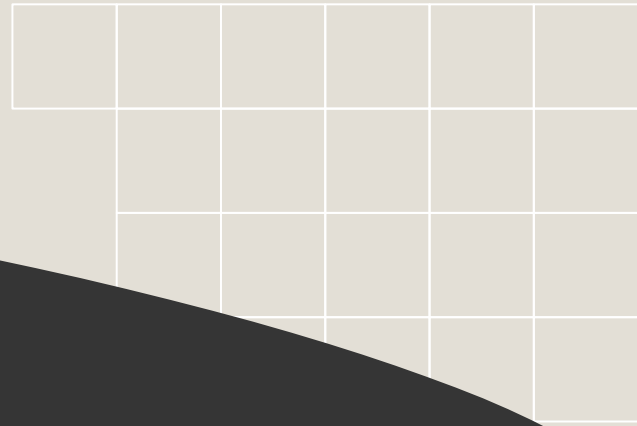
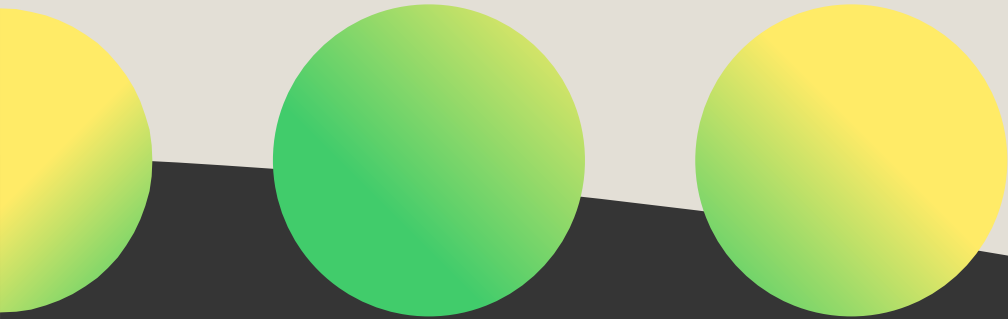
- **Baseline 1:** unconstrained least squares. This solution is the least parsimonious as all of our features are included in prediction.
- **LASSO (L1) regression** - least-squares objective function plus the L1 norm (sum of absolute values of coefficients), which shrinks coefficients to 0. Can result in more parsimonious models.
- **Ridge (L2) regression** - least-squares plus the L2 norm (sum of squares of coefficients), which shrinks coefficients but not necessarily to 0. Considered a middle-state.
- **Elastic net (L1 and L2) regression** - taken to incorporate both L1 and L2 norm terms to reduce and potentially eliminate features from the model.



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|                           | MSE                | RMSE               | MAPE                |
|---------------------------|--------------------|--------------------|---------------------|
| Base Model (Train)        | 649.1697103016952  | 25.478809044021176 | 0.142251819956421   |
| Base Model (Test)         | 1502.6547714880494 | 38.76409126353989  | 0.192792072449533   |
| L1 Model (Train)          | 648.7252821439231  | 25.47008602545196  | 0.1420531728498137  |
| L1 Model (Test)           | 1528.4750883869337 | 39.09571700822142  | 0.19442471415290927 |
| L2 Model (Train)          | 650.222949843282   | 25.49946959925406  | 0.1398374711514343  |
| L2 Model (Test)           | 1734.0362816915472 | 41.64176127028667  | 0.20334498640490548 |
| Elastic Net Model (Train) | 689.2256810795373  | 26.253108027042003 | 0.1411604886637268  |
| Elastic Net Model (Test)  | 1835.3355818880975 | 42.84081677428778  | 0.20683866965326528 |

# Key Insights



# Implementation 1 Discussion

***What are the key observations you made from the solutions as applied to the problem?***

- If features are engineered from prior exercise, diet, and glucose levels and patient id is incorporated as well, accuracy is ~9% more accurate (i.e., elastic net performance) than if patient id were not used.

***What does your solution mean for the problem?*** We compare and contrast our two model variants for Implementation 1.

- However, the model incorporating patient id as a feature may be valuable more so for existing patients when you're seeking to track patient health over time, rather than for new/prospective patients when data is unavailable or very limited.

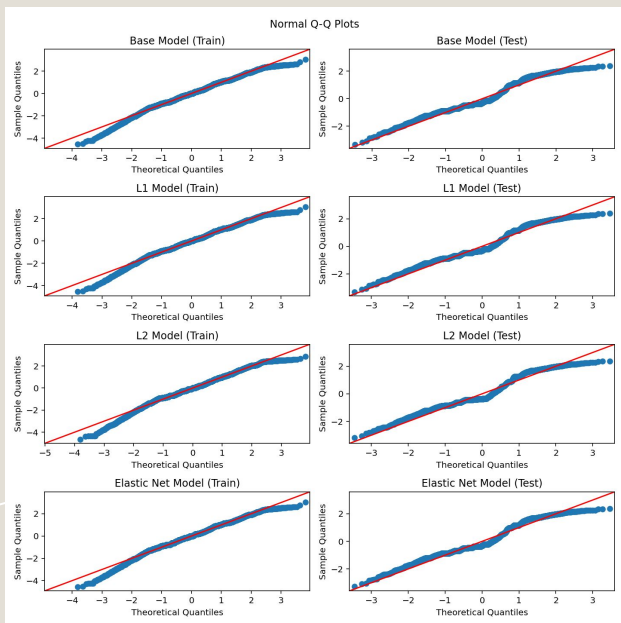




# Implementation 2 Discussion

## *What does your solution mean for the problem?*

- We compare and contrast our two model variants for Implementation 2:
- A conventional least-squares approach tradeoff for simplicity with more involved feature engineering.

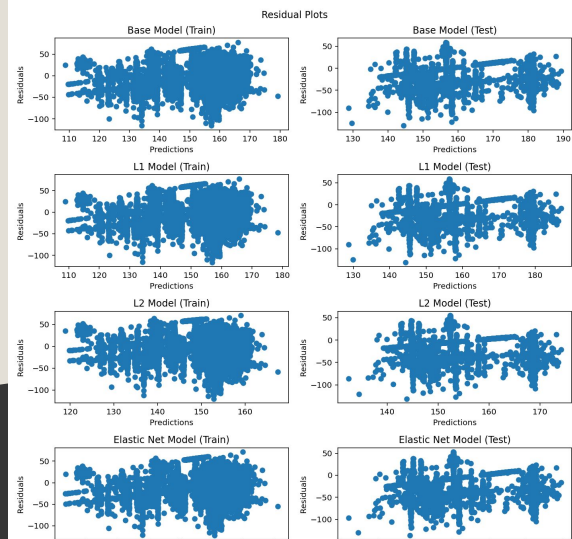


## *What are the key observations you made from the solutions as applied to the problem?*

- We examine residual plots as resulting from each of our models' predictions on both train and test samples, and note generally good fit, however, slight potential multi-collinearity. (slight uptrend in residual plots).

## *Any Limitations/When can these models be used?*

- A more simplistic approach if a patient does not have extensive health history (e.g., new/prospective patient)
- Can be used to establish a rough baseline for more advanced predictive models / elaborate feature engineering approaches.



# Major Conclusions

## ***Was your approach viable?***

- The exercise of exploiting two meaningfully different optimization approaches was valuable as either approach may be sufficient based on the desired level of sophistication for the modeler / health administration.
- A simpler, less feature engineered approach may be desired either for simple baselining of patients or for new/prospects, whereas a more nuanced model can better served for established patient tracking over time.

## ***Did you run into any difficulties such as infeasibility, numerical instability, unexpected results, etc.***

- Data ingestion was a key deliberation in our process, as we did not have traditional time series (i.e., gaps).
  - As a result, the simpler implementation, while objectively lower error, may likely incorporate more bias than the more complex feature-engineering, while increased variance.
- 

# Suggestions for Improvement and Future Work

***If you had more time and resources what would you have done to solve the problem?***

- Acquired more data (either contacted the author of the data set for additional samples, or sought expanded data sets on diabetes patients)
- We could also explore a baseline time series model (ie., autoregressive) or a Long-Short Term Memory Neural Network for time series forecasting. Given that true patient data likely tends to incorporate seasonality/cyclicity, these alternative models can potentially fit signal better than traditional least squares regression.

